FLAGSHIP 05 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-PCM/G

Flame Retardant PCM Cooling Vest · Antistatic & Flame-Retardant PCM Armo

Antistatic & Flame-Retardant PCM Armor. Passive Cooling for Hazardous Frontlines.

Engineered with high-energy-density Phase Change Material absorbing heat at constant ~23°C and encapsulated within an antistatic, flame-retardant outer shell. Designed for petrochemical, foundry, welding, and utility operations where spark hazards, explosive dust, and extreme heat converge.

01 Flame-Retardant & Antistatic Protection

Heavy-duty industrial twill fabric repels slag, sparks, and static charge accumulation.

02 Constant 22.08°C Thermal Regulation

Maintains continuous core cooling without cold shock during heavy manual labor in 35°C+ ambient heat.

03 High-Stability C17 Biomass Core

Thousands of thermal cycles without degradation; non-toxic and eco-friendly.

04 Night & Low-Light Visibility

Dual high-visibility silver reflective bands provide 360° safety in roadworks and dark industrial plants.

LC Frost-PCM/G · OFFICIAL HARDWARE VIEW



KEY METRIC 01
22.08°C
Non-Irritating Phase Change Comfort Zone

KEY METRIC 02
225 J/g
Latent Heat Storage Capacity

KEY METRIC 03
360°
High-Visibility Retro-Reflective Safety Bands

KEY METRIC 04
0V
Zero-Spark Intrinsically Safe Passive Cooling

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



FR & ANTISTATIC INDUSTRIAL TWILL

Antistatic & Flame-Retardant Ripstop Outer Shell

Specially treated antistatic and flame-retardant industrial weave dissipates electrostatic charges and resists welding splatter, enabling safe core cooling inside petrochemical refineries and gas plants.



360° RETRO-REFLECTIVE TRIM

Dual 360° Retro-Reflective Safety Warning Bands

Wide silver retro-reflective bands wrap 360° around the chest and waist, ensuring immediate operator recognition in low-light refineries, tunnels, and night-shift roadworks.



MODULAR PCM BAYS · 0.96KG TOTAL

Modular Internal PCM Bays & Elastic Waistband

Internal thermal compartments allow crews to swap freshly recharged 22.08°C C17 PCM packs in seconds while the wide magic-tape waistband locks the vest closely to the torso.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **FR & Antistatic Outer Weave [Flame-Retardant Industrial Twill Shell]**
   Rugged ripstop fabric resists tearing, abrasions, and molten metal splatter.
2. **Elastic Magic-Tape Waistband [Heavy-Duty Cummerbund Fastener]**
   Wide industrial hook-and-loop cummerbund fits securely across all torso builds.
3. **Modular PCM Thermal Pockets [Internal C17 Thermal Pack Bays]**
   Dedicated internal compartments hold sealed PCM packs tight against the core.
4. **Dual Safety Warning Reflective Bands [360° Retro-Reflective Warning Strips]**
   3M-grade retro-reflective stripes across the chest and waist ensure field visibility.

FIELD DEPLOYMENT | EX-RATED PETROCHEMICAL CRACKING UNIT | HIGH-HEAT ZONE



Petrochemical Refineries, Welding Bays & Night-Shift Utilities

In explosion-proof (EX) petrochemical zones and active welding shops, electronic batteries and brushed motors pose unacceptable ignition risks. The LC Frost-PCM/G pairs 0V passive 22.08°C phase-change cooling with an antistatic, flame-retardant outer shell and 360° reflective bands - delivering uncompromising thermal relief where electronics are forbidden.

0V Intrinsically Safe	22.08°C (225 J/g)	0.96kg
Ignition & Spark Hazard	Phase Change Comfort	Complete Vest Weight

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-PCM/G (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-PCM/G
IDENTITY & COMPLIANCE	**Color**	Industrial Grey with High-Visibility Reflective Trim (customizable)
IDENTITY & COMPLIANCE	**Target Users**	Petrochemical, metallurgy, welding, power grid, and high-temperature night-shift workers
PERFORMANCE & MECHANICS	**Heat of Fusion**	225 J/g
PERFORMANCE & MECHANICS	**Phase Change Temperature**	22.08°C
PERFORMANCE & MECHANICS	**Cooling Duration**	1-2 hours (reference based on labor intensity and ambient temp)
PERFORMANCE & MECHANICS	**Cold Storage Recharge Time**	Frozen at -18°C for 1 hour, OR refrigerated at 0-4°C for 1-2 hours
MATERIALS & ERGONOMICS	**Fabric Material**	Antistatic and flame-retardant fabric
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	0.96kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-PCM/G

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware